



Correlation Analysis in Traffic Noise Measurement (Case Study: Makassar City, Indonesia)



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Abstract: This study aims to analyze the traffic noise levels at three locations in Makassar City and to compare them with the established noise quality standards. Measurements were conducted over a one-week period at specific times using a sound level meter, a vehicle speed measurement device, and a counting application to classify vehicle types into heavy vehicles (HV), light vehicles (LV), and motorcycles (MC). The observation sites included an educational area, a hospital area, and a residential area. Correlation analysis using Statistical Package for the Social Sciences (SPSS) was employed to examine the relationships between HV, LV, MC, and vehicle speed with the equivalent continuous sound level (Leq). The results indicated that noise levels at all three locations exceeded the standard threshold of 55 decibels (dB). The correlation analysis showed significant relationships between Leq and HV (0.834), LV (0.782), MC (0.787), and vehicle speed (-0.680). The effective contribution to noise was highest for HV (40.44%), followed by MC (13.35%), LV (12.68%), vehicle speed (10.38%), and other factors (23.15%), including human activity, construction noise, road surface type, road gradient, and surrounding environmental conditions. Recommended mitigation measures include restricting the operating hours and rerouting of HV in sensitive areas, as well as enforcing noise emission testing and regulations on illegal exhaust modifications.

Keywords: Heavy vehicles; Light vehicles; Motorcycles; Speed, Noise Leq; Correlation analysis; Makassar City

1 Introduction

Traffic noise in urban areas is a form of pollution that is often overlooked, despite its significant impact on human health and the environment. With increasing urbanization and mobility, noise levels in major cities continue to rise, leading to numerous problems for residents living near major roadways.

Makassar City, the capital of South Sulawesi Province, has experienced rapid growth in recent years [1]. Resulting in increased traffic volume and noise. Traffic noise has various major impacts on human health. One study conducted by Akil et al. [2], found that noise levels in educational areas in Makassar City reached 76.44 A-weighted decibels (dB(A)), exceeding the quality standard of 55 dB(A) (Ministry of Environment Decision No. 48 of 1996). Noise can be a challenge for students to focus on. Moreover, hearing loss can result from long-term exposure to high noise levels. Increased risk of heart disease, stress, and sleep disorders, as well as psychological problems such as anxiety and depression. Sleep disturbances caused by traffic noise can result in decreased productivity and quality of life for the community. In addition, people living in areas with high levels of noise pollution often experience impaired concentration and decreased learning ability, indicating that noise can negatively impact the overall learning process. Therefore, researching and understanding how noise affects health are the first steps in designing better policies to protect the community [3].

The traffic characteristics in Makassar City, characterized by significant vehicle volume and traffic density, contribute to high noise levels. This condition certainly has an impact on reducing the quality of life of the surrounding community. Research on traffic noise in this city is vital not only to understand the existing noise levels, but also to develop effective solutions to address these issues.

In addition, according to Morillas et al. [4], traffic noise studies are important for improving urban planning. By understanding noise patterns across different urban areas, the government can identify locations that require intervention, such as the installation of noise barriers, the planting of vegetation for noise absorption, or the

implementation of traffic regulation adjustments. The data obtained from such studies can also be used to design more sustainable transportation systems, for example by reducing dependence on private motor vehicles through the development of more efficient and environmentally friendly public transport policies.

The purpose of this study is to measure traffic noise levels at several strategic locations in Makassar City, to compare the results with established noise quality standards, and to analyze the correlation between noise sources and observed noise levels. It is expected that the findings of this research will provide valuable information for future researchers, the city government, and the community. This information can serve as a basis for formulating appropriate policies and mitigation measures to reduce the impact of traffic noise and improve the quality of life of the people of Makassar.

Through this research, it is expected that awareness of the importance of traffic noise management will increase, as well as the need for more proactive measures from authorities to create a healthier and more comfortable environment for residents living near major roadways.

2 Literature Review

Based on the characteristics of motor vehicles as defined in the 1997 Indonesian Road Capacity Manual [5–7], various types of motorized vehicles are classified into heavy vehicles (HV), light vehicles (LV), and motorcycles (MC). Each category has distinct characteristics and capacities within the traffic system, as described below:

(a) HVs are motor vehicles with more than four wheels, including buses, two-axle trucks, three-axle trucks, and combination trucks.

(b) LVs are four-wheeled motor vehicles with two axles, such as passenger cars, minibuses, pick-up vehicles, and small trucks.

(c) MCs are motor vehicles with two or three wheels, including MC, pedicabs, and other three-wheeled vehicles.

According to studies [8, 9], speed is a rate of travel expressed in kilometers per hour (km/h) and is generally divided into three types, namely average speed, instantaneous speed, and maximum speed.

(a) Spot speed is a term commonly used in traffic analysis to describe the speed of a vehicle at a specific point on a roadway at a particular time.

(b) Travel speed is the average speed of a vehicle representing how fast it travels along a route. This speed is calculated by dividing the length of the route by the travel time when the vehicle is in motion.

(c) Travel speed can also be defined as the effective speed of a vehicle, calculated by dividing the distance travelled between two locations by the total time required to complete the trip.

Regulations regarding noise in Indonesia are governed by the Minister of Environment Decree No. 48 of 1996 on Noise Level Standards. These standards establish the maximum permissible noise levels in different environments to prevent adverse impacts on human health and to maintain environmental comfort. Based on land use and environmental zoning, the allowable noise levels are adjusted according to the function of each area. This classification is presented in a noise quality standards table, which specifies threshold limits for various types of areas, including educational institutions, hospitals, and residential zones (Table 1).

Table 1. Quality standards

Number	Zoning/Activity Environment	Noise Level
1	Land Use	
	Housing/settlements	55 dB(A)
	Trade/services	70 dB(A)
	Offices/commerce	65 dB(A)
	Green open space	50 dB(A)
	Industry	70 dB(A)
	Government/public facilities	60 dB(A)
	Recreations	70 dB(A)
	Special	
	a. Airport	
	b. Railway station	
	c. Port	70 dB(A)
	d. Cultural heritage	60 dB(A)
2	Activity Environment	
	Hospital	55 dB(A)
	Education	55 dB(A)
	Place of worship	55 dB(A)

According to Table 1, the standard noise quality limits established by the Ministry of Environment of the Republic of Indonesia specify that the permissible noise level for educational, residential, and hospital areas is 55 dB(A).

3 Research Methodology

This study analyzed the sources of noise in relation to established noise level quality standards based on land use suitability and examined the relationship between noise sources and measured noise levels. Noise measurements were conducted using a sound level meter, which, according to study [10], is an effective method for obtaining primary data on environmental noise levels. The device could measure sound within a range of 30–130 decibels (dB) and frequencies between 20 and 20,000 Hz, thereby providing an accurate representation of sound intensity from various sources, including vehicular noise. The number of vehicles passing the observation points was determined using a counting method similar to that applied in a previous study [11]. Meanwhile, vehicle speed was measured using a speed gun, as discussed in study [12]. The combination of noise measurements with traffic volume and vehicle speed data provided a more comprehensive understanding of the impact of traffic on environmental noise levels.

The study was conducted over a one-week period on selected days, namely Monday, Friday, Saturday, and Sunday (Table 2). Data collection took place at three locations representing educational, hospital, and residential areas.

Table 2. Selection of days

Days	Considerations for Selecting a Date
Monday	To represent normal weekday traffic conditions, just like Tuesday, Wednesday, and Thursday.
Friday	Describing a workday with religious activities and shorter working hours.
Saturday	Showing a weekend pattern combining work and leisure activities.
Sunday	To represent holidays with low mobility as a comparison to other days.

Table 2 presents the considerations for selecting observation days to capture variations in traffic noise levels. Monday is selected to represent typical weekday traffic conditions, like those on Tuesday, Wednesday, and Thursday. Friday represents a workday with distinct characteristics, including midday religious activities and shorter working hours. Saturday illustrates a weekend pattern that combines work-related and leisure activities, while Sunday represents a holiday with generally lower mobility, serving as a comparative baseline for other days.

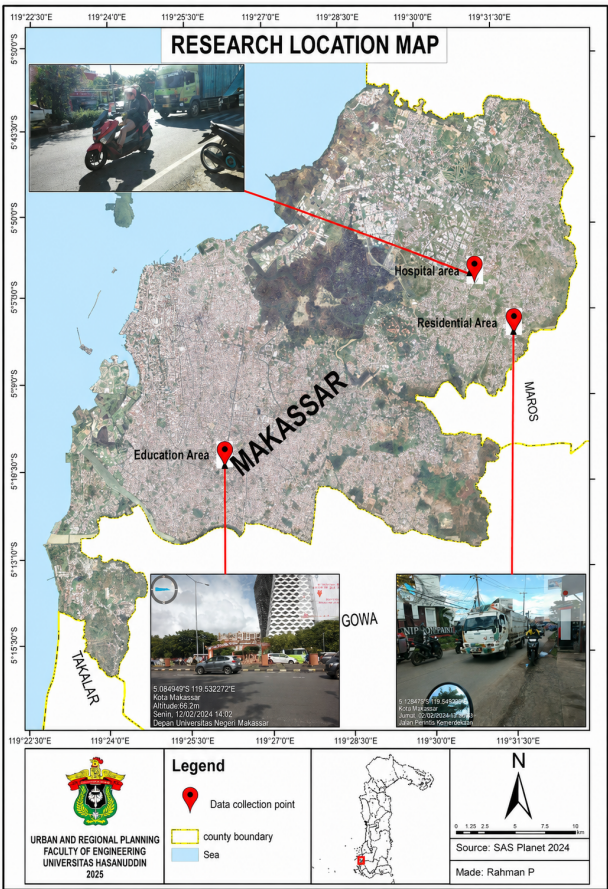


Figure 1. Research location map

Figure 1 shows the locations of the research sites for the educational, hospital, and residential areas. The educational site is located at the State University of Makassar on Andi Pangeran Pettarani Road. Data collection in the hospital area is conducted at Daya Hospital on Perintis Kemerdekaan Road. In addition, the residential site is situated in the Grand Rahmani Residence area on Paccerakang Street.

Table 3 presents data for three roads: Paccerakang Street (residential area), Perintis Kemerdekaan Road (hospital area), and Andi Pangeran Pettarani Road (educational area). The table includes information on road width, road length, and data collection locations.

Table 3. Data collection locations

Street Name	Geometric Conditions of Roads		Data Collection Location
	Road Width (m)	Road Length (m)	
Paccerakang street	6.34	4,132	Residential
Perintis Kemerdekaan street	22.84	12,100	Hospital
Andi Pangeran Pettarani street	38.66	4,300	Education

Paccerakang Street has a width of 6.34 meters and a length of 4,132 meters. Perintis Kemerdekaan Road is 22.84 meters wide and 12,100 meters long. Meanwhile, Andi Pangeran Pettarani Road has a width of 38.66 meters and a length of 4,300 meters. Overall, the table provides a clear overview of the characteristics of each road segment and the corresponding data collection points.

Maps that illustrate research locations across various sectors serve as essential visual tools for researchers. By presenting thematic information, these maps enable researchers to quickly identify and understand the geographical distribution of data collection sites, including educational institutions, healthcare facilities, and residential neighbourhoods.

This type of visual representation enhances the understanding of location points, enabling researchers to make informed decisions regarding the focus of their studies. It is particularly valuable in fields such as public health, geodesy, cartography and remote sensing, geomatics, geography, surveying, and regional and urban planning, where the context of data collection can significantly influence research outcomes.

Based on Appendix II of Decree No.48/MENLH/XI/1996 regarding noise level quality standards. The following are direct measurement methods:

(a) With reference to Minister of Environment Regulation No. 07 of 2009 regarding noise limits for new types of motor vehicles, the sound level meter must be positioned at a distance of 7.5 meters from the edge of the highway to ensure measurement accuracy.

(b) Based on KEP No. 48/MENLH/11/1996, the height is between 1.2 meters to 1.5 meters, and measurements were taken for 10 minutes at a time. Data were read every five seconds, resulting in 120 data points in a single measurement session.

(c) Referring to SNI 8427:2017 established by the national standardization agency, noise measurements were taken over a 24-hour period, representing specific times of day: 00:00–03:00 (01:00), 03:00–06:00 (04:00), 06:00–09:00 (07:00), 09:00–14:00 (10:00), 14:00–17:00 (15:00), 17:00–22:00 (20:00), 22:00–00:00 (23:00).

In data analysis, frequency distribution is an important tool for understanding the distribution of data. The process of constructing a frequency distribution begins with calculating the range, which is defined as the difference between the maximum and minimum values. This range is calculated using the formula ($r = \text{Data Max} - \text{Data Min}$). Once the range has been determined, the next step is to calculate the number of classes to be used in the frequency distribution. One commonly used method for determining the number of classes is the Sturges formula, ($k = 1 + 3.3 \log(n)$), as reported by Koutsoyiannis [13], where n is the number of data points. Once the number of classes is determined, the next step is to calculate the class interval width. The class interval is obtained by dividing the range r by the number of classes k . After defining the class intervals, the midpoint of each class can be calculated. This midpoint is determined by adding the lower limit (LL) and upper limit (UL) of each class interval and then dividing the result by two, as expressed by the formula $\text{Midpoint} = (\text{LL} + \text{UL})/2$. Through these steps, a frequency distribution can be constructed that includes class intervals, midpoints, and frequencies, thereby facilitating the analysis of Leq data.

The Leq formula, as proposed by Brink et al. [14], is used to calculate the average noise level over specific period, in this study defined as 10 minutes. This formula can be expressed as follows:

$$\text{Leq} = 10 \log \frac{1}{n} \sum T_n \cdot 10^{0.1L_n} \quad (1)$$

where,

Leq = Leq with sampling every 5 seconds during 10 minutes,

T_n = Frequency value,

L_n = Median value,

n = Amount of noise measurement data.

In statistical analysis, one method used to determine the relationship between two quantitative variables is the correlation test, which can be performed using the Statistical Package for the Social Sciences (SPSS) application [15]. The relationship between two variables may arise from a causal link or occur by chance.

According to Schober et al. [16], two variables are considered positively correlated if they change consistently in the same direction, while a negative correlation occurs when the variables change in opposite directions. In interpreting correlation test results, the criteria are based on the significance value (p -value). If the significance value is less than 0.05, it can be concluded that there is a statistically significant correlation between the two variables. Conversely, if the significance value is greater than 0.05, it indicates that no significant correlation exists.

4 Result and Discussion

4.1 Result

The results showed that the data on HV, LV, MC, as well as vehicle speed and noise levels, collected over one week on Monday, Friday, Saturday, and Sunday, provided a comprehensive overview of traffic conditions (in terms of volume and speed) and environmental noise at the study sites.

4.1.1 Vehicle volume and speed

The vehicle volume and speed survey were conducted over a one-week period, focusing on four representative days: Monday, Friday, Saturday, and Sunday, and covering three main area types: educational, hospital, and residential areas. The survey aimed to analyse traffic patterns and variations in vehicle speed based on area type and day of the week.

The collected data provided insights into how vehicle flow and speed varied depending on the functional characteristics of each area and the observation day, thus capturing the differences between weekday and weekend traffic patterns. These variations also reflected the influence of land use activities and temporal travel demand on overall traffic conditions in the study area.

Table 4. Vehicle volume and speed in educational areas

Location	Days	Time	Vehicle Volume			Speed (km/h)
			HV	LV	MC	
Andi Pangeran Pettarani Street	Monday	07:00	8	189	536	24.33
		10:00	11	197	553	21.67
		15:00	10	173	532	30.33
		20:00	8	162	438	30.67
		23:00	6	103	339	46.67
		01:00	3	72	171	56
		04:00	1	59	165	62.33
	Friday	07:00	5	211	603	19
		10:00	11	186	516	28
		15:00	10	188	548	26.67
		20:00	8	154	327	32.67
		23:00	4	110	298	48.67
		01:00	2	81	178	54.33
		04:00	5	96	282	51.67
	Saturday	07:00	6	131	358	38
		10:00	5	150	440	33.33
		15:00	9	174	478	28.67
		20:00	6	194	504	23.67
		23:00	3	125	343	46.67
		01:00	5	71	182	58
		04:00	1	57	135	60
	Sunday	07:00	7	130	398	38.67
		10:00	5	148	397	34.67
		15:00	8	143	508	37.33
		20:00	8	126	350	39.67
		23:00	5	111	304	43.33
		01:00	0	88	286	41.67
		04:00	2	100	291	52

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

Table 4 shows that Traffic conditions on Andi Pangeran Pettarani Street show clear variations depending on the time and day of observation.

Vehicle volume, particularly MC, peaks during the day and in the afternoon on weekdays, especially on Mondays and Fridays, reflecting heavy urban activity and movement. Conversely, traffic volume decreases significantly late at night and in the early morning, resulting in higher average vehicle speeds. Observations on weekends, particularly on Saturday and Sunday evenings, still show relatively high traffic activity, although speeds tend to be more stable compared to peak hours on weekdays.

Table 5 shows that traffic characteristics on Paccerakkang Road indicate a relatively lower vehicle volume compared to major arterial roads, with MC remaining the dominant vehicle type throughout the observation period.

Table 5. Vehicle volume and speed in residential areas

Location	Days	Time	Vehicle Volume			Speed (km/h)
			HV	LV	MC	
Paccerakkang Street	Monday	07:00	6	28	236	17.33
		10:00	4	44	160	27.67
		15:00	5	50	207	25
		20:00	3	35	73	46.33
		23:00	2	23	95	41
		01:00	1	12	33	57.33
		04:00	0	15	27	67
		07:00	4	61	207	23.33
	Friday	10:00	4	48	166	26.33
		15:00	6	44	223	15
		20:00	3	72	71	47.33
		23:00	2	12	62	50.67
		01:00	1	7	39	56.67
		04:00	0	5	37	60.5
		07:00	3	68	113	33.33
		10:00	4	57	98	36.33
	Saturday	15:00	6	37	137	29.33
		20:00	4	30	132	30.33
		23:00	1	20	102	35.33
		01:00	0	14	29	65.5
		04:00	1	7	30	56.33
		07:00	4	39	122	32.33
		10:00	4	28	82	45
		15:00	3	46	219	20
	Sunday	20:00	2	30	79	46.33
		23:00	4	22	54	53
		01:00	1	19	43	58.33
		04:00	2	14	58	49

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

Traffic density is generally higher during the day, particularly in the morning and afternoon on weekdays, which aligns with lower average speeds due to increased mobility activity. Conversely, vehicle volume decreases significantly during late-night and early-morning hours, leading to a significant increase in traffic speed, particularly between 1:00 a.m. and 4:00 a.m. Weekend traffic patterns appear more moderate and stable, reflecting reduced travel intensity compared to weekdays.

Table 6 shows that traffic conditions on Jalan Perintis Kemerdekaan consistently exhibit high traffic volume, dominated primarily by MC, reflecting its role as one of the main urban transportation corridors in Makassar. During the daytime on weekdays, particularly on Mondays and Fridays, traffic density is very high and accompanied by relatively low vehicle speeds, reflecting congestion and heavy mobility activity.

All of the tables above present the results of measurements and calculations, including the number of HV, LV, and MC, as well as the average vehicle speed. These data provide a clear overview of traffic conditions and serve as a

basis for further analysis.

Table 6. Vehicle volume and speed in hospital areas

Location	Days	Time	Vehicle Volume			Speed (km/h)
			HV	LV	MC	
Perintis Kemerdekaan Street	Monday	07:00	11	264	744	13.33
		10:00	6	195	727	14.33
		15:00	6	253	742	14.33
		20:00	8	185	710	16.33
		23:00	6	173	639	20.33
		01:00	4	83	285	51.33
		04:00	3	83	302	47.33
	Friday	07:00	9	223	621	26.67
		10:00	7	196	686	15.33
		15:00	7	257	692	17.67
		20:00	4	133	498	30.33
		23:00	2	115	470	39.33
		01:00	1	87	294	48.33
		04:00	1	86	238	55.67
	Saturday	07:00	7	141	316	43.67
		10:00	8	154	474	32.33
		15:00	7	177	580	32.33
		20:00	5	191	617	27.67
		23:00	5	129	366	44.33
		01:00	3	64	189	55.33
		04:00	2	75	82	56.67
	Sunday	07:00	5	159	432	34.33
		10:00	7	153	551	27.33
		15:00	10	194	638	23.67
		20:00	8	149	677	21.33
		23:00	5	112	567	31.67
		01:00	5	87	296	49.67
		04:00	2	96	313	48.33

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

4.1.2 Calculation of Leq value

The noise measurements conducted on Monday at 07:00 a.m. represented the time interval from 06:00 a.m. to 09:00 a.m., with each measurement lasting 10 minutes. Using a sound level meter, a total of 120 noise data points were recorded for each observation point and time interval.

The noise data were processed to determine the number of observations (n), maximum value (Max), minimum value (Min), range (R), number of classes (k), class interval (i), and the Leq value. The results obtained from each observation point were then analyzed using the following formulas.

$$\begin{aligned}
 \text{Amount of data } (n) &= 120 \\
 \text{Maximum value (Max)} &= 103 \\
 \text{Minimum value (Min)} &= 58 \\
 \text{Value range } (R) = \text{Max} - \text{Min} &= 103 - 58 = 45 \\
 \text{Count the number of classes } (k) &= 1 + 3.3 \log(n) \\
 &= 1 + 3.3 \log(120) \\
 &= 7.86 \sim 8 \\
 \text{Calculate class intervals } (i) &= r/k \\
 &= 45/8 \\
 &= 5.625 \sim 6
 \end{aligned}$$

After determining the range (R), number of classes (k), and class interval (i), a frequency distribution table was constructed as the basis for calculating the Leq, thereby ensuring that the noise measurement data were systematically organized to facilitate accurate analysis and interpretation. Table 7 presents the frequency distribution, illustrating the distribution of traffic noise level data in the educational area.

Table 7. Noise data observation points educational area Monday at 07:00

Class Interval	Median Value	Frequency	Frequency (%)
58–63	60	6	5
64–69	66	22	18.33
70–75	72	28	23.33
76–81	78	31	25.83
82–87	84	18	15
88–93	90	10	8.33
94–99	96	4	3.33
100–105	102	1	0.83
Totals		120	100

The data are grouped into eight class intervals, ranging from 58–63 dB to 100–105 dB. Each class interval has a midpoint value, which represents the estimated average noise level within that range, while the frequency indicates the number of observations recorded in each interval. In addition, the frequencies are expressed as percentages of the total number of observations, which amounts to 120 data points. After determining the class intervals, midpoint values, and frequencies, the calculation of the Leq is subsequently performed.

$$\begin{aligned}
\text{Leq (10minutes)} &= 10 \log \frac{1}{n} \sum T_n \cdot 10^{0.1 L_n} \\
&= 10 \log \left(\frac{1}{120} \right) (T_i \cdot 10^{0.1 \cdot L_i}) + \dots + (T_j \cdot 10^{0.1 \cdot L_j}) \\
&= 10 \log \left(\frac{1}{120} \right) [(6 \cdot 10^{0.1 \cdot 60}) + (20 \cdot 10^{0.1 \cdot 66}) \\
&\quad + (29 \cdot 10^{0.1 \cdot 72}) + (31 \cdot 10^{0.1 \cdot 78}) \\
&\quad + (18 \cdot 10^{0.1 \cdot 84}) + (11 \cdot 10^{0.1 \cdot 90}) \\
&\quad + (4 \cdot 10^{0.1 \cdot 96}) + (1 \cdot 10^{0.1 \cdot 102})] \\
&= 10 \log \left(\frac{1}{120} \right) [52504062793] \\
&= 86.41 \text{ dB(A)}
\end{aligned}$$

The calculation of the Leq value, measured over a 10-minute period, is a method used to describe sound intensity in dB. Noise observations were conducted in the educational area at 07:00 to obtain representative data on noise levels in the area.

The same calculation procedure was applied at all observation points and predetermined times, resulting in consistent and comparable Leq values. The collected data were then presented in graphical form, providing a visual representation of variations in noise levels across different locations and observation times.

These data not only describe the sound intensity at specific times but also facilitate the analysis of noise patterns that may occur in the area over time. The following figures present the Leq values for one week across the three data collection areas.

4.1.3 Leq noise value

The presented data show the Leq at the educational location over four days—Monday, Friday, Saturday, and Sunday—with measurements taken at predetermined time intervals. Noise levels are expressed in dB and reflect the variations in environmental noise experienced at the site.

Figure 2 shows that noise levels tend to be higher in the morning, particularly at 07:00 a.m., which may be attributed to increased early morning activities. Conversely, noise levels decrease significantly at night, especially after 08:00 p.m., indicating a reduction in activity at the educational site during nighttime hours.

Further analysis indicates that Monday exhibits the highest noise level, with a peak of 86.41 dB recorded at 07:00 a.m., which may reflect increased activity at the beginning of the week. In contrast, Saturday shows the lowest noise levels, with measurements reaching 68.49 dB at 04:00 a.m., indicating that the educational environment tends to be quieter at late night hours, particularly around 11:00 p.m. and between 01:00 a.m. and 04:00 a.m.

From Figure 3, it can be observed that noise levels tend to be higher during the daytime, particularly during peak hours such as 07:00 a.m. and 08:00 p.m. For example, on Saturday, noise levels peak at 15:00 with a value of 87.24 dB, indicating increased activity in the vicinity of the hospital at that time. Conversely, noise levels decrease significantly at night, with the lowest value recorded at 04:00 a.m. on Friday, with a Leq of 68.35 dB. These variations in noise levels may be influenced by traffic activity surrounding the hospital area. In a hospital environment, elevated

noise levels can negatively affect the comfort of both patients and staff. Therefore, noise monitoring and management are essential to maintain the overall quality of the healthcare environment.

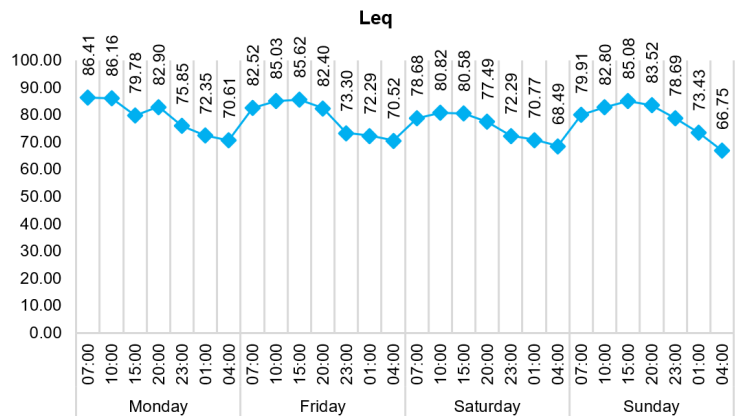


Figure 2. Leq values in educational areas

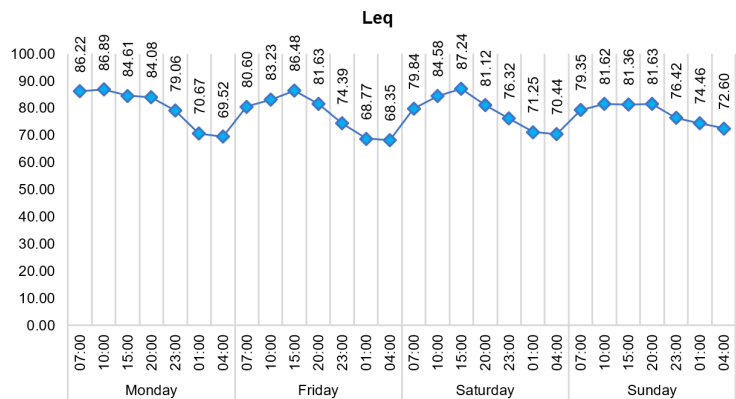


Figure 3. Leq value in the hospital areas

The presented data show the Leq in the hospital areas at various times and on different days of the week. The presented data also show the Leq in residential areas over several days of the week, with measurements taken at predetermined time intervals. Noise levels are expressed in dB and reflect the variations in environmental noise experienced by residents in the area.

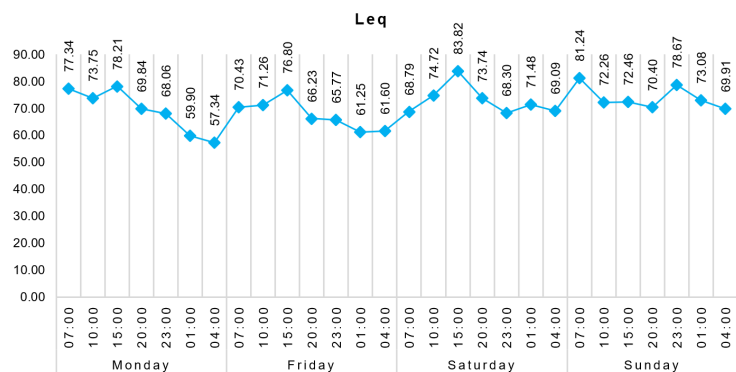


Figure 4. Leq values in residential areas

Figure 4 shows that the highest noise level occurred on Saturday at 15:00, with a value of 83.82 dB, while the lowest level was recorded on Monday at 04:00, with a value of 57.34 dB. This indicates that noise levels in residential areas tend to be higher on weekends, which may be attributed to increased social and recreational activities. Further

analysis of the data reveals variations in noise patterns throughout the day. During the early morning hours, noise levels tend to be lower, with the lowest values observed between 01:00 a.m. and 04:00 a.m. As the day progresses, noise levels increase, reaching a peak in the afternoon, before gradually decreasing again at night. This pattern is likely influenced by the daily activities of residents, as well as the relatively quieter conditions during nighttime hours.

4.2 Discussion

4.2.1 Comparison of noise quality standards with Leq values

The comparison between noise quality standards and the Leq values is essential for assessing the impact of noise on the environment. The noise standard, set at 55 dB, represents the maximum permissible threshold to maintain environmental comfort and prevent disruption to human activities. In contrast, the Leq represents the average noise level measured over a specific period, such as one hour or one day.

By comparing the measured Leq values with the established standards, it can be determined whether the noise levels at a given location remain within acceptable limits or exceed thresholds that may cause disturbances. If the Leq value exceeds 55 dB, the area can be classified as noisy, with the potential to negatively affect the health and comfort of the surrounding community.

Figure 5 presents the Leq in the educational, hospital, and residential areas, compared with the noise quality standard of 55 dB, based on measurements taken at various times on selected days, namely Monday, Friday, Saturday, and Sunday. The results indicate that all measured noise levels exceed the established quality standard of 55 dB, primarily due to noise-generating activities such as vehicular traffic.

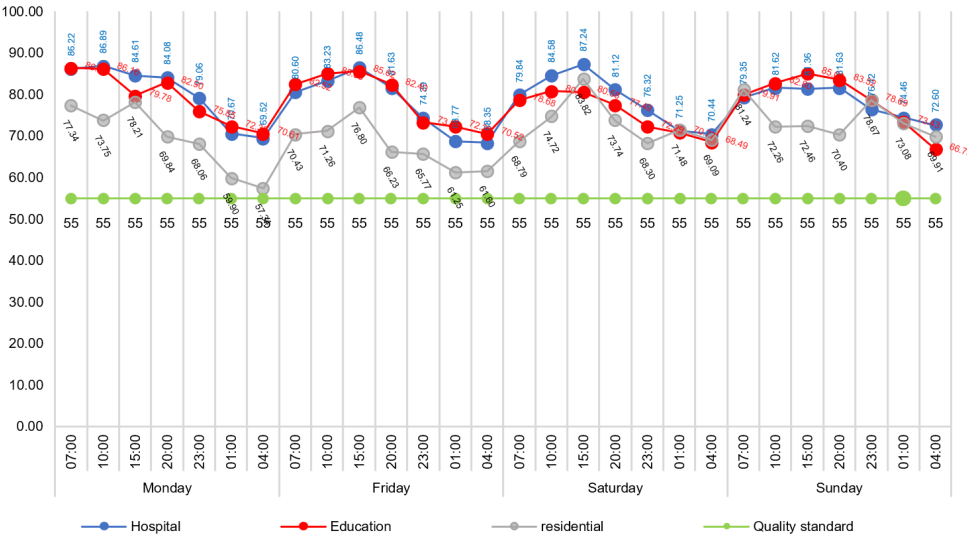


Figure 5. Comparison chart of noise quality standard with Leq value

The measurement results further show that noise levels in the educational area exhibit significant fluctuations throughout the day. The highest noise level occurs in the morning, particularly on Monday at 07:00, with a Leq value of 86.41 dB. In contrast, the lowest noise level occurs at night, specifically on Saturday at 04:00, with a Leq value of 66.49 dB. Although this value is lower than the morning measurements, it still exceeds the established noise quality standard. Studies [17, 18] indicated that noise levels in educational areas exceed the recommended limits. Similarly, research [19] showed that educational areas located near traffic corridors experience noise levels above the established quality standards, which can lead to impaired concentration among students.

For noise levels in the hospital area, the measurements indicate that the highest Leq occurred on Saturday at 15:00, reaching 87.24 dB. A similar pattern is observed on Monday, Friday, Saturday, and Sunday between 07:00 and 20:00, during which the Leq values frequently exceed 80 dB. In contrast, noise levels tend to decrease at night after 23:00, with the lowest values recorded at 01:00 and 04:00. The minimum Leq value was observed on Friday at 04:00, with a value of 68.35 dB. However, the Leq values in the hospital area remain above the established quality standard, indicating that disturbances to environmental tranquillity persist. According to research [20], noise levels around hospital areas exceed the recommended limits and disrupt environmental tranquillity. Similarly, Lakawa et al. [21] reported that traffic noise in the vicinity of hospitals surpasses acceptable health thresholds.

The data presented in the graph demonstrate the Leq in residential areas over several days of the week, with measurements taken at various time intervals. Leq represents the average noise level expressed in dB over a specified period.

As illustrated in the graph, noise levels vary depending on the time of measurement. For example, on Saturday at 15:00, the highest noise level was recorded at 83.82 dB, which exceeds the established quality standard. This indicates that at certain times, particularly when human and vehicular activities increase, noise levels in the area can become problematic.

In contrast, during early morning hours, such as 01:00 and 04:00, noise levels tend to decrease, with the lowest value reaching 57.34 dB on Monday at 04:00. However, this value still exceeds the prescribed noise quality standard. According to studies [22, 23], the measured noise levels exceed the recommended limits, thereby negatively affecting the health, communication, and comfort of occupants.

4.2.2 Classical assumption tests

The purpose of the Classical Assumption Tests is to verify whether a linear regression model satisfies the fundamental assumptions required for valid and reliable statistical inference.

(a) Normality test

To determine whether the obtained data are normally distributed, statistical analysis is performed. Normal distribution is a fundamental assumption in many statistical methods, including regression analysis and correlation tests. By conducting a normality test, researchers can evaluate the suitability of the data for the applied statistical model, thereby ensuring that the analysis results are more valid and reliable. The normality test (Table 8) includes several statistical indicators and the p -value obtained from the Kolmogorov-Smirnov test, each providing information on the extent to which the data deviate from a normal distribution.

Table 8. Normality test

		HV	LV	MC	Speed	Leq
N		84	84	84	84	84
Normal Parameters ^{a,b}	Mean	4.7143	105.119	323.0952	37.3288	75.8751
	Std. Deviation	2.86471	68.34769	214.20999	13.34267	6.97794
Most Extreme Differences	Absolute	0.091	0.087	0.091	0.083	0.084
	Positive	0.091	0.087	0.091	0.07	0.073
	Negative	-0.065	-0.071	-0.083	-0.083	-0.084
Test Statistic		0.091	0.087	0.091	0.083	0.084
Asymp. Sig. (2-tailed)		0.081 ^c	0.171 ^c	0.079 ^c	0.200 ^{c,d}	0.200 ^{c,d}

Note: a. Test distribution is Normal; b. Calculated from data; c. Lilliefors Significance Correction; d. This is a lower bound of the true significance. HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

According to the Kolmogorov-Smirnov test results, several significance values need to be considered. For the HV, LV, and MC variables, the significance values are 0.081, 0.171, and 0.079, respectively. Although these values are above the 0.05 threshold, they are relatively close to the significance limit, indicating that the data may deviate slightly from a normal distribution.

Meanwhile, for the speed and noise variables, the significance value for each is 0.200, which is well above 0.05. This indicates that there is insufficient evidence to reject the null hypothesis of normality. Therefore, the data for these variables can be considered normally distributed, providing confidence for subsequent analyses under the assumption of normality.

Overall, the analysis of significance values from the Kolmogorov-Smirnov test provides important insights into the data distribution. Although some variables show potential deviations from normality, others satisfy the normality assumption, thereby supporting the validity of further statistical analyses.

(b) Linearity test

Data analysis aimed to determine whether there is a significant linear correlation between the independent variables and the dependent variable. Before conducting further analysis, it is essential to ensure that the data meet the assumption of linearity, so that the results of the analysis can be interpreted appropriately and reliably.

Table 9 presents the results of the linearity significance test for several variables, namely HV, LV, MC, and vehicle speed. The significance values obtained for each variable are notably low, with HV, LV, and speed having values of 0.000, indicating that their relationships with the dependent variable are highly significant at the 0.05 level. Meanwhile, the MC variable has a significance value of 0.006, which also indicates a statistically significant relationship, although slightly higher than those of the other variables.

The high level of significance across all variables indicates the presence of strong and consistent relationships between the independent variables and the dependent variable. This finding is important in the context of data analysis, as it suggests that variations in HV, LV, MC, and vehicle speed have a significant influence on the dependent variable. Therefore, a deeper understanding of these relationships can support more informed decision-making and the development of more effective strategies in relevant applications.

Table 9. Linearity test

Variable	Significance	Deviation from Linearity	Significance value	Description
HV	0.000	0.077	0.05	Linear
LV	0.000	0.061	0.05	Linear
MC	0.006	0.568	0.05	Linear
Speed	0.000	0.165	0.05	Linear

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

(c) Correlation test

The statistical method used to evaluate and measure the strength and direction of the relationship between two variables is known as the correlation test. This test aims to determine the extent to which two variables are related, either positively or negatively. However, before conducting correlation analysis, it is essential to ensure that the data meet certain assumptions, such as normality and linearity. The normality test ensures that the data distribution follows a normal pattern, while the linearity test verifies that the relationship between variables is linear.

If these assumptions, normality and linearity, are satisfied, the results of the correlation test can be considered more valid and reliable, thereby supporting accurate conclusions regarding the relationship between the variables.

Table 10 presents the results of the correlation test conducted to evaluate the relationships between several variables, namely HV, LV, MC, and vehicle speed, with the Leq noise level. The analysis results indicate that the HV variable has a correlation coefficient of 0.834, demonstrating a very strong positive relationship with noise levels.

Table 10. Correlations test

		HV	LV	MC	Speed
	Pearson Correlation	0.834**	0.782**	0.787**	-0.680**
	Sig. (2-tailed)	0.000	0.000	0.000	0.000
Leq	Sum of Squares and Cross-products	1383.39	30966.87	97651.68	-5252.75
	Covariance	16.67	373.09	1176.53	-63.29
	N	84	84	84	84

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

This finding indicates that HV contributes the most to traffic noise. Therefore, future policies should address traffic management related to HV operations within urban areas, particularly regarding operating hours. This result implies that an increase in the number of HV significantly contributes to higher noise levels, indicating that areas with high HV volumes tend to experience elevated noise levels.

Consequently, government authorities and relevant stakeholders can prioritize targeted interventions, for example by regulating HV operations as the primary source of noise. This correlation also supports the development of data-driven policies, such as restricting operating hours, rerouting HV away from sensitive areas, and establishing low-noise zones.

In addition, the LV and MC variables also showed positive relationships with noise levels, with correlation coefficients of 0.782 and 0.787, respectively. Although these relationships are not as strong as that observed for HV, both variables still exhibit strong correlations with noise levels. This indicates that an increase in the volume of LV and the number of MCs in an area contributes to higher noise levels, although their impact is less pronounced than that of HV.

These findings are important for understanding the factors influencing noise levels in urban environments, particularly in the context of traffic planning and management. Moreover, the presence of strong correlations can support technical and administrative arguments in the formulation of new regulations or the revision of existing policies. For example, recognizing that MC and LV also contribute significantly to noise pollution allows for the broader implementation of policies such as sound emission testing and restrictions on exhaust modifications.

Overall, these correlation results enable the development of more targeted and efficient noise control policies, which can directly contribute to improving the quality of life of communities, particularly those residing in noise-sensitive areas such as residential zones, schools, and hospitals.

Overall, the strong relationships between the HV, LV, and MC variables and noise levels indicate a significant association that cannot be overlooked. This consistent positive correlation confirms that increases in these variables correspond to higher levels of environmental noise, and vice versa. In other words, as the volume of vehicles increases, the noise level also rises. This finding is consistent with previous study [24], which reported that traffic volume is directly proportional to noise levels.

In contrast, the speed variable shows a negative relationship with noise levels, with a correlation coefficient of -0.680. This indicates that higher vehicle speeds are associated with lower noise levels. This negative relationship

may occur because vehicles traveling at higher speeds tend to produce more uniform and continuous noise, whereas slower-moving vehicles in congested traffic conditions generate greater noise due to frequent acceleration, deceleration, and engine load variations. This finding is consistent with studies by Lu et al. [25] and Ibili et al. [26], which reported that vehicle speed is inversely related to noise levels. However, other research conducted by Barros et al. [27], indicates a contrasting result, stating that higher vehicle speeds are associated with increased noise levels.

5 Calculation of Effective Contribution

An analysis of effective contribution was conducted to determine the percentage of influence of each independent variable (X) on the dependent variable (Y). This calculation uses the Pearson correlation coefficient (r) and the beta regression coefficient (β) from the results of the multiple linear regression test. The Pearson correlation value indicates the degree of relationship between the independent variables and the dependent variable, while the beta value indicates the magnitude of each variable's influence in the regression model. The analysis results show that the HV variable has the highest correlation value and beta coefficient, thus providing the greatest effective contribution to the noise level compared to the other variables.

In Table 11, the β values indicate the magnitude of the influence of each independent variable on the dependent variable after the data has been standardized. In the analysis of effective contribution, the β values are used in conjunction with r to determine the contribution of each variable to the noise level. The regression results show that the HV variable has the highest β value of 0.485, making it the most dominant factor influencing traffic noise. The MC and LV variables have moderate effects with beta values of 0.170 and 0.162, respectively. Meanwhile, the speed variable has a negative β value of -0.153, indicating an inverse relationship with noise levels.

Table 11. Coefficients

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	β		
(Constant)	69.763	2.362		29.541	0.000
1	HV	1.181	0.485	5.361	0.000
	LV	0.017	0.162	0.899	0.371
	MC	0.006	0.170	0.914	0.363
	Kecepatan	-0.080	-0.153	-1.885	0.063

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

Based on Table 12, the multiple correlation coefficient (R) of 0.877 indicates that the relationship between the independent variables and noise levels is very strong. The R^2 value of 0.768 indicates that 76.8% of the variation in noise levels can be explained by the variables HV, LV, MC, and speed.

Table 12. Model summary

Model	R	R^2	Adjusted R^2	Std. Error of the Estimate
1	0.877	0.768	0.757	3.44161

Based on Table 13, the data require the calculation of effective contribution. After applying the effective contribution formula, the variables that contribute most significantly to noise levels can be identified as follows:

Table 13. Effective contribution calculation

Variable	Coefficient Regression (β)	Coefficient Correlation (r)	Amount	Coefficient of Determination (R^2)
HV	0.485	0.834	0.404459	0.768
LV	0.162	0.782	0.126722	
MC	0.170	0.787	0.133504	
Speed	-0.153	-0.680	0.103854	
Total			0.768538	

Note: HV, heavy vehicles; LV, light vehicles; MC, motorcycles.

Table 13 shows the magnitude of each variable's contribution to traffic noise levels based on the β , r , and contribution value (Amount). The HV variable has the most dominant influence on noise, with a β value of 0.485 and a r of 0.834, and provides the largest effective contribution of 0.404459, or approximately 40.45%. The LV variable also shows a strong relationship with noise, with an r value of 0.782 and a contribution of 0.126722, or approximately 12.67%. Furthermore, the MC variable has a β value of 0.170, a correlation of 0.787, and an effective contribution

of 0.133504, or approximately 13.35%. Meanwhile, the vehicle speed (Speed) variable has a negative regression coefficient of -0.153 and a correlation value of -0.680, indicating that an increase in vehicle speed tends to reduce noise levels under certain conditions, with a contribution of 0.103854 or approximately 10.39%. Overall, the R^2 value of 0.768 indicates that 76.8% of the variation in noise levels can be explained by the variables HV, LV, MC, and Speed, while the remainder is influenced by other factors outside the research model. To clarify Figure 6, this study has included a pie chart showing the percentage contribution of each variable to the level of traffic noise.

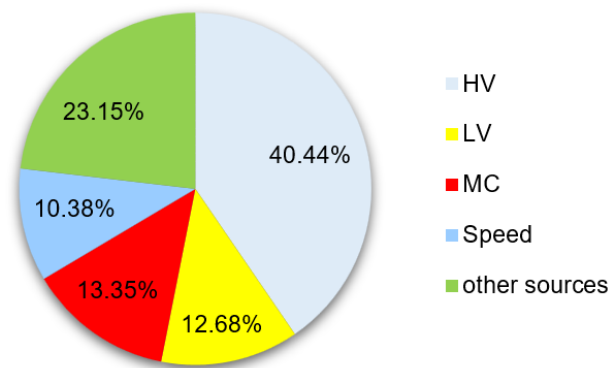


Figure 6. Percentage diagram of sources that affect noise

The high levels of traffic noise in urban areas such as Makassar are influenced by various types of vehicles with different technical characteristics, with HV identified as the primary contributor, accounting for 40.44%. Factors such as large engine capacity, deteriorated exhaust systems, and heavy loads increase the intensity of noise generated by HVs, particularly when traveling on uneven or rough road surfaces. Therefore, the control of HV operations should be prioritized, for example through restricting operating hours during peak periods and rerouting traffic away from sensitive areas.

Meanwhile, LV and MC also contribute significantly to noise due to their overwhelming presence on roads. Technical control through the implementation of regular sound emission tests is crucial to ensure noise levels do not exceed the thresholds set in Ministry of Environment and Forestry Regulation No. 56 of 2019. For example, MC with a capacity of >175 cc have a maximum limit of 83 dB(A), but test results show that some vehicles reach noise levels of 87.3 dB(A), typically due to the use of modified exhaust systems (racing). This has the potential to cause serious disturbances in densely populated areas. Therefore, in addition to enforcement measures such as fines or administrative sanctions, public education through campaigns promoting the use of standard exhaust systems and collaboration with the automotive community are necessary to create a more orderly and healthy acoustic environment in the long term.

6 Conclusions

Based on the results of Leq in three sensitive zones—educational areas, hospital areas, and residential areas—it was found that all locations exhibited noise levels exceeding the established quality standard of 55 dB. The highest Leq value was recorded in the hospital area (87.24 dB), followed by the educational area (86.41 dB), and the residential area (83.82 dB). This indicates that traffic activity, particularly during peak hours, significantly contributes to increased noise levels, which may negatively affect comfort and overall public health over time.

Correlation test results indicate a very strong and statistically significant relationship between vehicle volume (HV, LV, and MC) and noise levels. The HV variable shows the strongest correlation with Leq ($r = 0.834$), followed by MC and LV. In contrast, vehicle speed exhibits a negative correlation ($r = -0.680$), suggesting that higher speeds are associated with lower noise levels, possibly because vehicles experience fewer stopping and acceleration cycles.

The analysis of effective contributions further supports these findings, with HV contributing 40.44% to total noise, making it the dominant factor. MC and LV contribute 13.35% and 12.68%, respectively, while vehicle speed contributes 10.38%. The remaining 23.15% is attributed to other sources, such as human activities, construction work, and road conditions.

As vehicle usage continues to increase each year, traffic noise is likely to intensify. Therefore, noise control in urban areas such as Makassar should be implemented comprehensively, including regulating HV operations, enforcing noise emission standards, and adopting both educational and technical approaches, in order to create a healthier and more comfortable acoustic environment.

Author Contributions

Conceptualization, R.P. and V.V.N.; methodology, R.P.; software, R.P.; validation, R.P., V.V.N., and Y.K.D.S.; formal analysis, R.P. and V.V.N.; investigation, R.P.; resources, R.P.; data curation, R.P., V.V.N., and Y.K.D.S.; writing—original draft preparation, R.P., V.V.N., and Y.K.D.S.; writing—review and editing, R.P. and V.V.N.; visualization, R.P.; supervision, R.P. and V.V.N.; project administration, R.P. and V.V.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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